

Epileptic Seizure Suppression by Focal Brain Cooling With Recirculating Coolant Cooling System: Modeling and Simulation

Kei Hata, Koichi Fujiwara^{ID}, *Member, IEEE*, Takao Inoue, Takuto Abe, Takatomi Kubo, *Member, IEEE*, Toshitaka Yamakawa^{ID}, *Member, IEEE*, Sadahiro Nomura, Hirochika Imoto, Michiyasu Suzuki, and Manabu Kano^{ID}, *Member, IEEE*

Abstract—A focal brain cooling system for treatment of refractory epilepsy that is implantable and wearable may permit patients with this condition to lead normal daily lives. We have developed such a system for cooling of the epileptic focus by delivery of cold saline to a cooling device that is implanted cranially. The outflow is pumped for circulation and cooled by a Peltier device. Here, we describe the design of the system and evaluate its feasibility by simulation. Mathematical models were constructed based on equations of fluid dynamics and data from a cat model. Computational fluid dynamics simulations gave the following results: 1) a cooling device with a complex channel structure gives a more uniform temperature in the brain; 2) a cooling period of <10 min is required to reach an average temperature of 25.0°C at 2 mm below the brain surface, which is the target temperature for seizure suppression. This time is short enough for cooling of the brain before seizure onset after seizure prediction by an intracranial electroencephalogram-based algorithm; and 3) battery charging would be required once every several days for most patients. These results suggest that the focal brain cooling system may be clinically applicable.

Index Terms—Computational fluid dynamics, epilepsy, focal brain cooling, Pennes bioheat equation, titanium.

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K. Hata, T. Abe, and M. Kano are with the Department of Systems Science, Kyoto University, Kyoto 615-8085, Japan.

K. Fujiwara is with the Department of Systems Science, Kyoto University, Kyoto 615-8085, Japan, and also with the Department of Material Engineering, Nagoya University, Nagoya 464-8601, Japan (e-mail: fujiwara.koichi@material.nagoya-u.ac.jp).

T. Inoue and M. Suzuki are with the Department of Neurosurgery, School of Medicine, Yamaguchi University, Ube 753-8511, Japan.

T. Kubo is with the Department of Information Science, Nara Institute of Science and Technology, Ikoma 630-0192, Japan.

T. Yamakawa is with the Department of Computer Science and Electrical Engineering, Kumamoto University, Kumamoto 860-8555, Japan.

S. Nomura and H. Imoto are with the Department of Neurosurgery, School of Medicine, Yamaguchi University, Ube 753-8511, Japan, and also with the Epilepsy Center, Yamaguchi University Hospital, Ube 755-0046, Japan.

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I. INTRODUCTION

EPILEPSY refers to a group of chronic disorders that are associated with excessive neuronal activity in the cerebrum and characterized by recurrent seizures that may manifest as convulsion, consciousness disturbance, or an unusual sensation. Epileptic seizures increase the risk of injury and affect social activities of patients, including a limitation or prohibition against acquiring a driving license in many countries.

Epileptic patients constitute about 1% of the population, and in 25% current treatments do not relieve seizures [1], [2]. Anti-epileptic drugs eliminate seizures in two-thirds of patients and are generally the first treatment option [3]. Surgery is used for patients who do not respond to these drugs, but the only surgical option is resection of the epileptic focus with high neuronal activity. This cannot be achieved for an epileptic focus located in the superficial cortical surface because of the difficulty of surgery for neocortical epilepsy involving a functional area [4]. Such cases are treated by palliative surgical options such as callosotomy, vagus nerve stimulation, and responsive neurostimulation, but these procedures do not eliminate seizures. This background indicates the need for a new approach to treatment of epilepsy.

Focal brain cooling involves application of hypothermia to the epileptic focus. Evidence for the efficacy of this method includes the suppression of stimulation-evoked seizures in intraoperative brain mapping within 5–10 s by pouring cold Ringer's lactate solution over the surface of the stimulated cortex [5]. In addition, interictal epileptiform activity can be suppressed by treatment of the brain surface with cold saline close to the epileptic focus [6].

Cooling of layers II and III in the gray matter underlies the suppressive effect of focal brain cooling [7], with a target temperature in this region of 25–30 °C [8]. This target temperature was confirmed in several clinical experiments in patients with epilepsy. Therapeutic whole-body hypothermia under 33° has a preventive effect on status epilepticus [9]. Fujii *et al.* [10] reported that epileptic discharges of EEG diminished during cooling when the brain surface temperature

reached less than 25°C. Seizures in a patient with refractory epilepsy were suppressed when the brain temperature was cooled to 28°C or 30°C by focal brain cooling under an awake condition [11]. Sourek and Travnicek [12] tested focal brain cooling below 24°C in 15 patients, and found that seizures were reduced in 11 patients and unaltered in four.

To cool these layers to 25 °C, a device applied to the brain surface needs to cool the surface temperature to <25 °C. The temperature of the brain surface may need to be 10-20°C, depending on the individual and the target region for cooling [8], [13], [14]. Application of penicillin G solution or cobalt powder to the brain surface in rats, followed by placement of a Peltier cooling device on the application area, caused frequent seizures if the device was not activated. Less frequent seizures occurred in rats with brain surfaces cooled to 20 °C or 15 °C by the Peltier device, and those with brain surfaces at 10 °C had no seizures [13]. An epileptic electroencephalogram (EEG) discharge induced by kainic acid was also suppressed by cooling the brain surface to 14.8±1.5 °C with the Peltier device [8], and seizures induced by injection of 4-aminopyridine solution were significantly reduced in duration by cooling the brain surface to about 20 °C [14]. Contact of a cooling device with the brain surface at 0-10 °C does not cause irreversible side effects. Among rats in which the brain surface was maintained at 20, 15, 10, 5, 0, or -5 °C for an hour, only rats maintained at -5 °C had a reversible decrease in motor function and irreversible histological damage [15]. Long *et al.* [16] found that cooling of the Broca region and the speech motor cortex in awake neurosurgical patients affected speech timing and articulation. Impairment in finger movements was observed when the brain surface of a monkey was cooled to 10 °C by a Peltier device, but the fingers were not completely paralyzed [17]. Two human patients had apparent temporal disruption of motor speech functions when focal hypothermia was applied to the Broca region with a stainless-steel chamber in which saline was circulating at 1-3 °C [18].

Several techniques for brain cooling have been proposed, including injection of cold fluid into blood vessels and covering patients with cooling blankets [19]. While these techniques reduce the body core temperature, it is unrealistic to lower the core temperature to 25 °C, even under anesthesia. Packing the head with ice can also be used, such that the steady-state brain surface temperature is about 20 °C; however, simulations under this condition suggest that the temperature in layers II and III would not be reduced to 25 °C [20].

A Peltier device provides one approach for cooling of the brain. This device generates a temperature difference of several tens of degrees Celsius between the hot and cold sides. Focal brain cooling systems using the Peltier device have been tested in clinical trials before resection of the epileptic focus, as well as in animal experiments. Fujii *et al.* [13] found that cooling the epileptic focus with an implantable Peltier device in direct contact with the brain suppressed epileptic seizures in a free-moving rat model. Nomura *et al.* [21] used a Peltier device for cooling the epileptogenic cortex in patients and found intraoperatively that epileptic discharges on electrocorticography (ECoG) were eliminated in the cooling

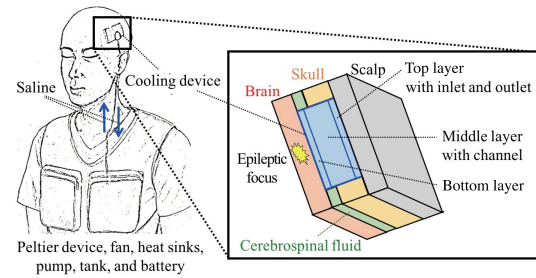


Fig. 1. Focal brain cooling system with RCC.

period and reappeared upon rewarming. The cortical GABA level also decreases significantly from the control level during cooling in patients [22]. Although the Peltier device can be used in a focal brain cooling system, safety is a concern in placing this device in direct contact with the brain because it is not biocompatible and a large electrical current is required for its function.

Recirculating coolant cooling (RCC) has been proposed to avoid these problems. The coolant is cooled outside the body, flows into a cooling device attached to the brain, removes heat from the brain, and is then cooled again outside the body. RCC permits adequate cooling and provides a therapeutic effect, based on studies by King *et al.* [56]. using a RCC brain cooling system in a traumatic brain injury monkey model, and development by Inoue *et al.* [23] of a RCC titanium cooling plate that covers the epileptic focus. Experiments in cat and monkey models showed that the titanium cooling plate cooled the focal cortical temperature sufficiently for seizure suppression. These results suggest that brain cooling with RCC is a promising therapy for epilepsy.

Our aim is clinical application of a RCC focal brain cooling system so that patients with refractory epilepsy can lead normal daily lives. Patients who would benefit from this system are those with an epileptic focus in the gyrus because this location is easy to cool with a device on the brain surface [4], while surgical treatment of neocortical epilepsy in a functional area (e.g., eloquent cortex) is difficult.

To achieve focal brain cooling with RCC, its feasibility has to be investigated through simulation before clinical trials with patients. In the present work, we developed a mathematical models for the whole brain cooling system. Based on the developed model, an optimal cooling device was designed from the perspectives of the pressure drop in the device and cooling performance, because the pump size increases as the pressure drop increases. The designed device was evaluated using computational fluid dynamics (CFD) simulations.

A preliminary version of this work has been described [24], but this report did not include the experiment in the cat model or the channel structure optimization in the cooling device.

II. BRAIN COOLING SYSTEM

Our RCC brain cooling system is shown schematically in Fig. 1. The patient wears a jacket with two chest pockets, one of which contains a Peltier device in contact with a tank of saline that is connected to a titanium cooling device implanted just above the epileptic focus in the skull. The circulating

saline is cooled by the Peltier device while in the tank, and then flows into the cooling device under pressure from a pump attached to the chest. The cooled saline removes heat from the brain and then flows back into the tank. Heat sinks are in contact with the cold and hot sides of the Peltier device, and this device and the pump are powered by a mobile battery that is also attached to the chest. The pump and battery are located extracorporeally and there is a risk of infection, but this kind of configuration is not uncommon in medical devices. For example, some types of ventricular assist devices have a pump and battery in this location [32], [33]. Over 2500 FDA-approved mechanical circulatory support devices (MCSs), including left ventricular assist devices (LVADs), were implanted in 2016 in the US [34], and the prevalence of infection was 19% at 12 months after implantation [35]. This shows that appropriate infection risk control is also needed in a brain cooling system with RCC.

Two cooling processes can be used: a seizure prediction-based operation and an intermittent operation. Intracranial EEG-based algorithms have been investigated for several decades [36]. In the American Epilepsy Society Seizure Prediction Challenge 2016, an area under the curve (AUC) value of 0.81 for long-term human data was achieved by the top algorithm [37]. In addition, a clinical trial of seizure prediction by intracranial EEG had been accomplished [38]. These algorithms can predict epileptic seizures 89 ± 15 min prior to their onset [39]–[41]; thus, the time for cooling the brain must be less than that from seizure prediction to onset. On the other hand, Yang *et al.* [42] proposed a method for cooling the epileptic focus intermittently like VNS and deep brain stimulation. Their experiments in a rat model showed that intermittent brain cooling reduces the frequency and intensity of epileptic seizures.

III. SYSTEM MODEL

This section describes a mathematical model of the RCC brain cooling system, which we refer to as the system model. Nomenclature for all symbols is given after the Conclusion.

A. Chest Devices

The patient wears a jacket with chest devices: a battery, fan, heat sinks, Peltier device, thermoregulator, and pump. The specifications are shown in Table I. These devices weigh just over 1 kg and occupy a volume of $120 \times 100 \times 70$ mm on each side of the chest. The two batteries are connected in series to raise the voltage, and the capacity is assumed to be 10 Ah, which is the standard capacity of mobile batteries in 2018 [25].

B. Model Structure

The main part of the system model is a hemispheric head model (Fig. 2), which comprises a titanium cooling device, saline, and biological tissue divided into soft tissue, skull, cerebrospinal fluid (CSF), and brain tissue [35]. The cooling device has a three-layer structure: the top layer has an inlet and an outlet for saline, the middle layer has a channel, and the bottom layer is a cooling panel in contact with the brain. The cooling device measures $30 \times 30 \times 7$ mm, and

TABLE I
SPECIFICATION OF DEVICES ON PATIENT'S CHEST

Device	Model number, Company	Weight (g), Size (mm)	Voltage (V), Current (A)
Two batteries	MPC-CW10000 [25], Maxell	180, $22 \times 63.5 \times 96$	5, Varied
Fan (hot side)	HY60W05 [26], Nippon Keiki Works	Not shown, $60 \times 60 \times 6$	5, 0.24
Heat sink (cold side)	EA(66)(H0404)(B) [27], Mizutani Electric Ind.	71, $70 \times 67 \times 12$	0, 0
Heat sink (hot side)	PIF-114-9-100-50 [28], Mizutani Electric Ind.	540, $114 \times 100 \times 50$	0, 0
Peltier device	FPH1-12708AC [29], Z-MAX	Not shown, $40 \times 40 \times 4$	Varied, Varied
Thermo regulator	sPLC-10 [30], Kurag Electronics	15, $40 \times 30 \times 18.5$	Varied, 0.1
Pump	CM-15-6 [31], Enomoto Micro Pump	60, $24 \times 31 \times 74$	6, 0.2

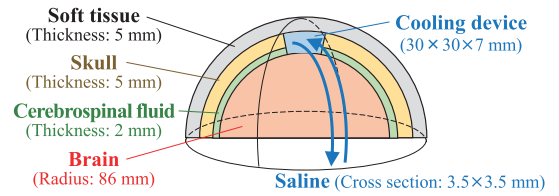


Fig. 2. Geometrical representation of head model.

TABLE II
CHANNEL STRUCTURE IN COOLING DEVICE. THE BLUE AND RED REGIONS REPRESENT THE INLET AND THE OUTLET OF THE SALINE, RESPECTIVELY

Device type	A	B	C
Channel structure			
Width of channels (mm)	13.5	3.83	1.9
Width of fins (mm)	1	1	1
Size of inlet and outlet (mm)	3.5×3.5	3.5×3.83	3.5×1.9

top, middle, and bottom layers are 1, 5, and 1 mm thick, respectively. Some types of epileptic lesions, such as focal cortical dysplasia (FCD), are localized within limited regions, and this device size is appropriate for FCD. Several kinds of device sizes should be considered depending on brain lesions.

Three types of channel structure in the middle layer are shown in Table II. The tubes for saline inflow and outflow are implanted under the soft tissue layer. Models of the titanium cooling device, saline flow, and biological tissue are described in sections III-C, III-D, and III-E, respectively.

C. Titanium Cooling Device

Temporal changes in the device temperature are calculated by the following heat conduction equation.

$$c_{Ti} \frac{\partial T}{\partial t} = k_{Ti} \nabla^2 T \quad (1)$$

D. Saline Flow

Temporal changes in the velocity, pressure, and temperature of saline are modeled as follows, assuming the flow to be

laminar:

$$\frac{\partial \rho_s}{\partial t} + \nabla \cdot (\rho_s \mathbf{u}) = 0 \quad (2)$$

$$\rho_s \frac{\partial \mathbf{u}}{\partial t} + \rho_s (\mathbf{u} \cdot \nabla) \mathbf{u} = \nabla \cdot [-p\mathbf{I} + \boldsymbol{\tau}] \quad (3)$$

$$\begin{aligned} \rho_s c_s \left(\frac{\partial T}{\partial t} + (\mathbf{u} \cdot \nabla) T \right) = & -\frac{T}{\rho_s} \frac{\partial \rho_s}{\partial T} \bigg|_p \left(\frac{\partial p}{\partial t} + (\mathbf{u} \cdot \nabla) p \right) \\ & - (\nabla \cdot \mathbf{q}) + \boldsymbol{\tau} : \mathbf{S} \end{aligned} \quad (4)$$

The temperature dependence of the material properties [43] is described by

$$\begin{aligned} \nu_s = & 1.38 \times 10^{-6} - 2.12 \times 10^{-8} T + 1.36 \times 10^{-10} T^2 \\ & - 4.65 \times 10^{-13} T^3 + 8.90 \times 10^{-16} T^4 \end{aligned} \quad (5)$$

$$\begin{aligned} \rho_s = & 839 + 1.40 T - 3.01 \times 10^{-3} T^2 \\ & + 3.72 \times 10^{-7} T^3 \end{aligned} \quad (6)$$

$$\begin{aligned} c_s = & 1.20 \times 10^4 - 80.4 T + 0.310 T^2 \\ & - 5.38 \times 10^{-4} T^3 + 3.63 \times 10^{-7} T^4 \end{aligned} \quad (7)$$

$$\begin{aligned} k_s = & -0.869 + 8.95 \times 10^{-3} T - 1.58 \times 10^{-5} T^2 \\ & + 7.98 \times 10^{-9} T^3. \end{aligned} \quad (8)$$

E. Biological Tissue

Temporal changes in the biological tissue temperature are calculated with a modified bioheat equation, which is based on the conventional bioheat equation proposed by Pennes [44]:

$$\rho_{tis} c_{tis} \frac{\partial T}{\partial t} = k_{tis} \nabla^2 T - c_{blood} w_{tis} (T - T_{core}) + M. \quad (9)$$

This equation represents energy conservation. The three terms on the right side represent the effects of heat conduction, heat convection due to blood flow, and metabolic heat production, in order. The equation assumes that the heat convection rate due to blood flow is proportional to the temperature difference between the tissue and the body core. Equation (9) has been verified in several studies [45], [46].

We derived the modified Pennes bioheat equation,

$$\rho_{tis} c_{tis} \frac{\partial T}{\partial t} = k_{tis} \nabla^2 T - c_{blood} W_{tis} (T - T_{core}), \quad (10)$$

with two modifications. The rate of production of metabolic heat was disregarded because this is much smaller than that of heat convection due to blood flow [47]. In addition, the blood perfusion rate w_{tis} was replaced by W_{tis} . The value of W_{tis} differs between the brain and other tissues (0 in CSF, skull, and soft tissue). W_{tis} in the brain was determined in an animal model to fit the theoretical temperature distribution to the experimental distribution. The details of this experiment are described in section. IV.

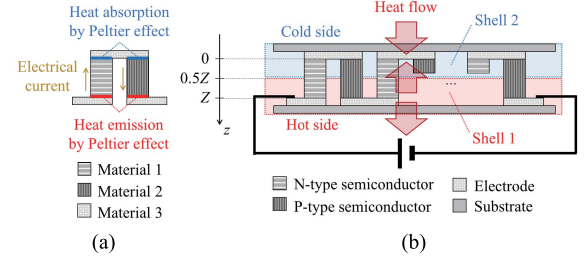


Fig. 3. Peltier device. (a) Basic unit. (b) Typical structure.

F. Initial and Boundary Conditions of Head Model

The boundary condition defines the outer surface of the head model as thermally insulated except for the saline inlet and outlet. The boundary condition for a steady state simulation is

$$T_{in,h} = T_{out,h} - \frac{Q_C}{F c_s \rho_s} \quad (11)$$

and that for an unsteady-state simulation is

$$T_{in,h}(t) = \begin{cases} T_{core}, & \text{for } t < \tau \\ T_{out,h}(t - \tau) - \frac{Q_C}{F c_s \rho_s}, & \text{for } t \geq \tau. \end{cases} \quad (12)$$

The term $-Q_C / F c_s \rho_s$ denotes the effect of heat absorption by the Peltier device. The Peltier effect and Peltier device are described in section III-G, and determination of the heat absorption rate is discussed in section III-H. The initial condition for the unsteady-state simulation is that the temperature of the whole head model is T_{core} .

G. Peltier Effect and Peltier Device

The Peltier effect is a phenomenon in which heat is absorbed or emitted at the contact surface of two materials through which an electrical current flows. When electrical current flows from material 1 to material 2, the heat absorption rate Q_C is expressed as

$$Q_C = (S_2 - S_1) T_{1,2} I. \quad (13)$$

The Peltier device is a plate device for cooling by the Peltier effect. The basic unit of the Peltier device is illustrated in Figure 3(a). For $S_1 < S_3 < S_2$ and electrical current flowing in the direction of the arrows, heat is absorbed on the top (blue) surfaces and emitted from the bottom (red) surfaces. A typical structure of the Peltier device is shown in Figure 3(b). Multiple basic units are connected in series two-dimensionally to increase the heat absorption rate. Materials 1 and 2 are thermoelectric chips, and material 3 is an electrode. The electrodes are in contact with substrates for electrical insulation. Heat absorption and emission caused by the Peltier effect result in a temperature drop around the top (blue) surfaces and a temperature rise around the bottom (red) surfaces. Hence, heat is conducted in the direction shown in Figure 3(b).

H. Heat Absorption Rate by Peltier Device

The Peltier device model calculates the heat absorption rate Q_C from a specified voltage U . The equations for energy

conservation of materials 1 and 2 (Fig. 3(b)) are as follows:

$$C_H \frac{dT_H}{dt} = -Q_H(t) + NST_H(t)I(t) - k_{\text{chip}}A \frac{\partial T}{\partial z} \Big|_{z=\frac{1}{2}Z} + \frac{1}{2}rI(t)^2 \quad (14)$$

$$C_C \frac{dT_C}{dt} = Q_C(t) - NST_C(t)I(t) + k_{\text{chip}}A \frac{\partial T}{\partial z} \Big|_{z=\frac{1}{2}Z} + \frac{1}{2}rI(t)^2 \quad (15)$$

The four terms on the right in each equation are the effect of heat emission from the hot/cold side, heat emission/absorption by the Peltier effect, heat conduction inside the Peltier device, and Joule heating, in order. There are two assumptions in Eqs. (14) and (15): the value of k_{chip} is the same for both thermoelectric chips, and the heat accumulation rate in materials 1 and 2 can be approximated as $C_H \frac{dT_H}{dt}$ and $C_C \frac{dT_C}{dt}$, respectively. The validity of Eqs. (14) and (15) has been shown previously [48]. Since the temperature gradient inside the thermoelectric chips is approximately linear along the z -axis [49], the following equation applies.

$$\frac{\partial T}{\partial z} \Big|_{z=\frac{1}{2}Z} = \frac{T_H(t) - T_C(t)}{Z} \quad (16)$$

Substitution of Eq. (16) for Eqs. (14) and (15) results in

$$C_H \frac{dT_H}{dt} = -Q_H(t) + NST_H(t)I(t) - \frac{k_{\text{chip}}A}{Z}(T_H(t) - T_C(t)) + \frac{1}{2}rI(t)^2 \quad (17)$$

$$C_C \frac{dT_C}{dt} = Q_C(t) - NST_C(t)I(t) + \frac{k_{\text{chip}}A}{Z}(T_H(t) - T_C(t)) + \frac{1}{2}rI(t)^2. \quad (18)$$

The Peltier device makes contact with heat sinks on the hot and cold sides. Thermal grease is applied for maximizing heat transfer between the device and the heat sinks. Equations (19) and (20) are satisfied based on the definition of heat resistance of the heat sink.

$$R_H Q_H(t) = T_H(t) - T_a \quad (19)$$

$$R_C Q_C(t) = T_{\text{out},h}(t) - T_C(t) \quad (20)$$

where R_H and R_C are overall heat resistances that express the sum of heat resistance between the Peltier device and the heat sinks, the heat resistance of the heat sinks themselves, and the heat resistance between the heat sinks and air or saline. In addition, the Kirchhoff circuit law results in

$$U(t) = rI(t) + NST_H(t) - NST_C(t). \quad (21)$$

Eqs. (17)-(21) define the Peltier device model. The initial conditions for the unsteady-state simulation are $T_H(0) = T_a$ and $T_C(0) = T_{\text{core}}$. The physical properties of the

device (NS , r , and $\frac{Ak_{\text{chip}}}{Z}$) are determined by the following equations [48].

$$NS = \frac{U_{\text{max}}}{T_H} \quad (22)$$

$$r = \frac{U_{\text{max}}(T_H - \Delta T_{\text{max}})}{I_{\text{max}}T_H} \quad (23)$$

$$\frac{Ak_{\text{chip}}}{Z} = \frac{I_{\text{max}}U_{\text{max}}(T_H - \Delta T_{\text{max}})}{2 T_H \Delta T_{\text{max}}} \quad (24)$$

I. Pressure Drop

To calculate the pressure drop during saline circulation, the whole system needs to be simulated; however, this requires heavy computation. A simple model of the relationship between the saline flow rate and the pressure drop was built to reduce the computational load.

The cooling device was divided into two parts: the inside of the middle layer (channel) and the outside (tube), which was approximated as cylindrical and of length 2 m. CFD simulations should be conducted with Eqs. (2)-(4) for calculating the pressure drop in the channel, but the pressure drop in the tube can be calculated as follows:

$$\Delta p_{\text{out}} = f \cdot \frac{\rho_s \bar{u}^2 L_t}{2d_t} \quad (25)$$

$$f = \begin{cases} 64 Re^{-1} & Re < 3000 \\ 0.0791 Re^{-0.25} & Re > 3000 \end{cases} \quad (26)$$

Use of the pressure drop model for the tube saves computational time in the CFD simulation.

IV. PARAMETER DETERMINATION IN A CAT MODEL

The principles, methods, and results of the experiment with a cat for determining the parameter W_{tis} in Eq. (10) are described. The thickness of the gray matter does not differ among animal species, but varies in different brain regions, and is about 2-4 mm overall [50]. This indicates that heat conduction in a cat's brain is close to that in humans. In addition, the body of a cat is large enough to prevent the body temperature becoming hypothermic during brain cooling. Thus, we used a cat for determining the parameter W_{tis} .

A. Principles

Let z be the coordinate perpendicular to the brain surface, with $z = 0$ on the brain surface and $z > 0$ inside the brain. Under the assumption that the brain temperature depends only on z , Eq. (10) is transformed to

$$\rho_{\text{tis}} c_{\text{tis}} \frac{\partial \tilde{T}}{\partial t} = k_{\text{tis}} \frac{\partial^2 \tilde{T}}{\partial z^2} - c_{\text{blood}} W_{\text{tis}} \tilde{T} \quad (27)$$

where $\tilde{T} = T - T_{\text{core}}$. The following equation is satisfied in the steady state.

$$\frac{d^2 \tilde{T}}{dz^2} = \frac{c_{\text{blood}} W_{\text{tis}}}{k_{\text{tis}}} \tilde{T} \quad (28)$$

The boundary conditions of Eq. (28) are as follows.

$$\tilde{T}(0) = \tilde{T}_s \quad (29)$$

$$\lim_{z \rightarrow \infty} \tilde{T}(z) = 0 \quad (30)$$

TABLE III
AGENTS INJECTED TO CAT IN EXPERIMENT

Active ingredient	Concentration (mg/mL)	Dose (mg/kg)
Ketamine	50	50
Atropine	0.5	0.05
Diazepam	5	2
Pentobarbital	64.8	35
Fentanyl	0.05	0.02

The solution of Eqs. (28)-(30) is

$$\tilde{T}(z) = \tilde{T}_s \exp\left(-\sqrt{\frac{c_{\text{blood}} W_{\text{tis}}}{k_{\text{tis}}}} z\right) \quad (31)$$

if none of c_{blood} , W_{tis} , and k_{tis} depend on z . This equation represents the steady-state temperature profile in the brain. It is transformed into

$$\ln \frac{\tilde{T}(z)}{\tilde{T}_s} = az \quad (32)$$

where the constant a is expressed as

$$a = -\sqrt{\frac{c_{\text{blood}} W_{\text{tis}}}{k_{\text{tis}}}}. \quad (33)$$

The value of W_{tis} is obtained through the following procedure.

- 1) Conduct an animal experiment in which the brain is cooled until reaching a steady state, and measure the steady-state temperature in the cooling device and at multiple depths in the brain.
- 2) Derive the constant a from the measurements $\ln(\tilde{T}(z)/\tilde{T}_s)$ vs. z by using least squares.
- 3) Determine W_{tis} as

$$W_{\text{tis}} \simeq \frac{k_{\text{tis}}}{c_{\text{blood}}} a^2. \quad (34)$$

Note that $c_{\text{blood}} = 4.2 \times 10^3 \text{ J/(kg} \cdot \text{K)}$ and $k_{\text{tis}} = 0.51 \text{ J/(m} \cdot \text{s} \cdot \text{K)}$ in the brain [51].

B. Methods

The experiment was performed on a cat weighing 4.0 kg. All procedures were reviewed by the Graduate School of Medicine Committee for Animal Experimentation at Yamaguchi University, and were performed based on the committee guidelines. First, five agents (Table III) were injected for general anesthesia, suppression of saliva, muscle relaxation, and analgesia. Since general anesthesia suppresses thermoregulation and respiration, an electric blanket was placed on the cat and a tube was inserted into the trachea. After tracheal intubation, the cat was maintained with controlled mechanical ventilation using anesthesia equipment (A.D.S. 1000, Engler Engineering Corp.). The head was then fixed with a stereotaxis apparatus and the skin incised to expose the skull. The dura was exposed by resecting part of the skull (Fig. 4 (a)).

The cooling device for measuring the brain temperature is shown in Fig. 5 (a) and (b). This device has a needle with three thermocouples. The needle was inserted into the brain perpendicular to the brain surface as shown in Fig. 6. The cooling device was pressed against the brain surface just above the needle, accompanied by two thermocouples at the inlet and

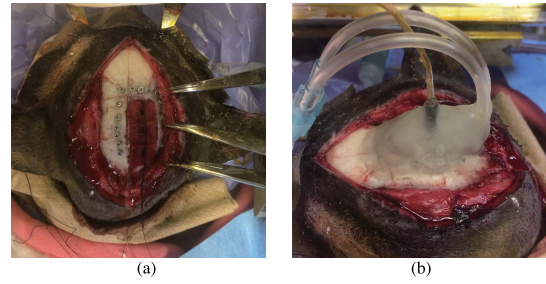


Fig. 4. Device implant area and actual cooling device. (a) Device implant area. (b) Cooling device fixed with resin.

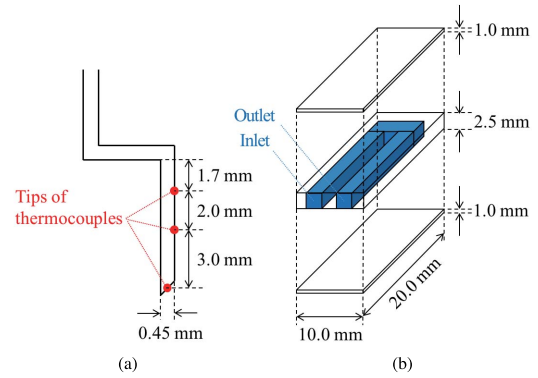


Fig. 5. Devices used in experiment. (a) Temperature measuring device. (b) Cooling device.

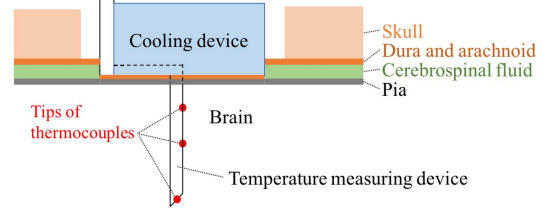


Fig. 6. Placement of temperature measuring device and cooling device.

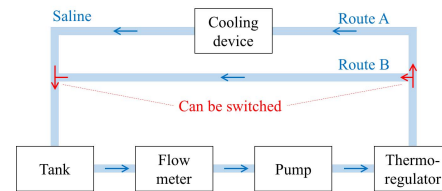


Fig. 7. Circulation of saline. There are two routes; only route A passed through cooling device.

outlet, and fixed to the skull with resin, after which the skin was sutured using a stapler (Fig. 4 (b)).

The two circulation routes for saline are shown in Figure 7. Both routes pass through a tank, a flow meter (LM05ZZT-AR, Horiba Tec), a thermoregulator (TTM-204, Toho Electronics and PSC-02, TwinBird), and a pump (KE-162, Taiko Kikai Industries), but only route A runs through the cooling device. Since the wall of the tubes was not thermally insulated, heat transfer between saline and air was not necessarily negligible.

After stopping the anesthesia equipment and waking the cat from anesthesia, brain temperature in the awake and naturally relaxed condition was recorded under the following conditions. First, saline was circulated in route B. The set point values

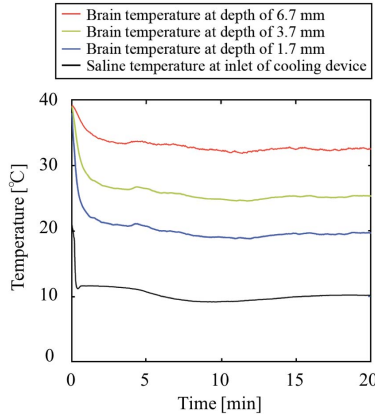


Fig. 8. Experimental results of cooling animal brain with saline of 10°C for 20 minutes.

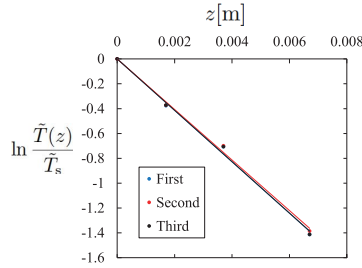


Fig. 9. The relationship between $\ln(\tilde{T}(z)/\tilde{T}_s)$ and z . The measurement data at three different depths are approximated by a straight line. A total of three lines are drawn on the basis of three animal experiments.

of the saline flow rate and the outlet temperature of the thermoregulator were 0.3 L/min and 8.5 °C, respectively, and the inlet temperature in the cooling device was about 10 °C. After the outlet temperature of the thermoregulator was decreased to the set point value, a cycle of 20 min cooling and 15 min rewarming was repeated three times. Cooling and rewarming correspond to use of routes A and B, respectively.

C. Results

Throughout the cooling period, the inlet temperature of the cooling device was maintained almost within the range of 10 ± 1 °C, and T_s was set to 10 °C in Eq.(32). The results of cooling for 20 min are shown in Figure 8. A time of <10 min was required for the brain temperature to level off. The steady-state brain temperature was approximated as the average temperature in the interval from 10 to 20 min after the start of cooling. T_{core} was set to 39.6 °C, which was the brain temperature at a depth of 6.7 mm after leveling off in the rewarming period.

A plot of $\ln(\tilde{T}(z)/\tilde{T}_s)$ vs. z is shown in Fig. 9. The slopes in the three cycles were -210 , -207 , and -204 m^{-1} . Based on the average slope, W_{tis} was determined to be 5.2 $kg/(m^3 \cdot s)$. The W_{tis} value was concluded to be valid because of the similarity of the slopes in different cycles.

V. SIMULATION

Simulation results for the RCC brain cooling system using the developed system model are described in this section. COMSOL Multiphysics® 5.1 was used for the simulation.

TABLE IV

COMPARISON OF CHANNEL STRUCTURES FOR PROBLEM 1

Device type	A	B	C
Temperature on bottom of cooling device (°C)			
Maximum	20.7	19.5	19.0
Average	17.8	17.8	17.8
Minimum	17.3	17.6	17.7
Temperature 2 mm below cooling device (°C)			
Maximum	29.4	29.0	28.9
Average	25.0	25.0	25.0
Minimum	24.2	24.3	24.3
Saline circulation			
Flow rate (L/min)	0.4	0.4	0.4
Pressure drop (kPa)	4	5	14

TABLE V

COMPARISON OF CHANNEL STRUCTURES FOR PROBLEM 2

Device type	A	B	C
Temperature on bottom of cooling device (°C)			
Maximum	19.9	19.7	19.6
Average	17.8	17.7	17.8
Minimum	17.6	17.4	17.3
Temperature 2 mm below cooling device (°C)			
Maximum	29.2	29.1	29.1
Average	25.0	25.0	25.0
Minimum	24.3	24.3	24.3
Saline circulation			
Flow rate (L/min)	1.0	0.1	0.06
Pressure drop (kPa)	20	2	1

A. Channel Structure in Cooling Device

The channel structure in the titanium cooling device requires optimization because it affects cooling performance and the pressure drop. Two types of optimization problems were solved to define the optimal structure of the channel among the candidates shown in Table II.

Problem 1: Minimize the difference between the highest and lowest temperatures on planes 0 and 2 mm below the cooling device at steady state. The constraints were as follows.

- An average temperature of 25.0 °C at 2 mm below the cooling device.
- A flow rate of saline of 0.4 L/min.

Problem 2: Minimize the pressure drop during saline circulation with the following constraints.

- An average temperature of 25.0 °C at 2 mm below the cooling device.
- A difference of 2.3 °C between the highest and lowest temperatures at the bottom of the cooling device at steady state.

The simulation results are shown in Tables IV and V. In problem 1, device C achieved the minimum difference between the maximum and minimum temperatures at planes 0 and 2 mm below the cooling device. In problem 2, device C also achieved the lowest pressure drop. Thus, device C has the optimal channel structure among the three candidates. In these simulations, a voltage of 2.0-2.1 V was required for Peltier device supply at steady state, and the hot and cold sides of the device had temperatures of <40 °C and >5 °C in both problems and all devices.

The brain surface temperature distribution 10 min after the start of cooling by device C is shown in Figure 10. The highest temperature at 2 mm below the cooling device did not reach the target temperature of 25 °C, even after cooling for 20 min.

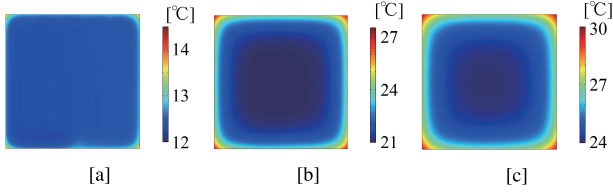


Fig. 10. Steady-state temperature distribution in brain for device C 10 minutes after beginning of cooling. (a) Bottom of cooling device. (b) 2 mm below cooling device. (c) 3 mm below cooling device.

The highest temperature occurred in the peripheral area of the device, which suggests that the cooling device should be larger than the target cooling region.

The device with the more complex structure achieved a temperature distribution with a higher uniformity. Vortexes occurred in the cooling device with a simple channel structure due to the wide channel width, and this hinders achievement of a uniform temperature distribution. In fact, as shown in Fig. 10, the highest temperature occurred in the periphery of the device, while the central region had the lowest temperature. Higher temperature uniformity is achieved when heat conduction from the periphery to the center is promoted. A channel structure with more titanium fins may be most effective, since titanium has a much larger thermal conductivity than saline at 25 °C: 21.9 W/(m · K) vs. 0.56 W/(m · K) (Table VII, Eq. (8)).

B. Cooling Performance

The cooling performance of device C was evaluated. The period required for brain cooling was estimated by unsteady-state simulation. The W_{tis} parameter in Eq. (10) was set to 5.2 kg/(m³ · s) (determined in section IV) and 10.4 kg/(m³ · s), which is double the determined value to simulate a situation in which the brain is more difficult to cool. The voltage for driving the Peltier device was fixed at 5 V, and the initial saline temperature was 38 °C.

As shown in Fig. 11, <10 min was required for the average temperature at 2 mm below the cooling device to reach the target of 25 °C for both values of W_{tis} . The hot and cold sides of the Peltier device were <40 °C and >5 °C, respectively, and the average electrical current over 20 min was 4.0 A.

Epileptic seizures can be predicted 89 ± 15 min before onset using an intracranial EEG-based algorithm that has a sensitivity of 91.3% and a false warning rate of 0.121 /h [39]–[41]. Thus, the time required for brain cooling is much shorter than that from seizure prediction to onset. Also, the period for brain cooling might be shorter than the calculated value if the initial saline temperature is <38 °C. This result shows that the cooling performance of the designed brain cooling device is sufficient for seizure suppression.

C. Frequency of Battery Charge

The frequency of battery charge is a key performance indicator for the focal brain cooling system because frequent charging or exchange of the battery is undesirable. The frequency of battery charge g is calculated as follows:

$$g = \frac{b}{\bar{I} t_{cool} \zeta} \quad (35)$$

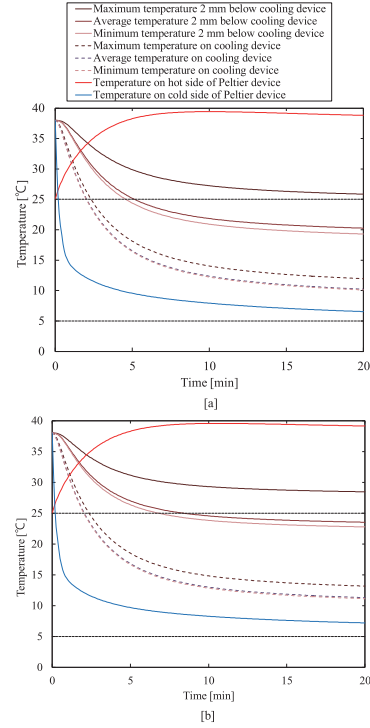


Fig. 11. Temporal change of temperature in brain and Peltier device when ambient temperature is 25 °C. (a) W_{tis} is 5.2 kg/(m³ · s). (b) W_{tis} is 10.4 kg/(m³ · s).

TABLE VI

VALUES OF CONSTANTS ON HEAD MODEL [51]

	Brain	CSF	Skull	Soft tissue
c_{tis} (J/kg·K)	3,630	4,096	1,313	3,391
k_{tis} (J/m·s·K)	0.51	0.57	0.32	0.37
ρ_{tis} (kg/m ³)	1,046	1,007	1,908	1,109

where b , $\bar{I}$, t_{cool} , and ζ represent the battery capacity, the average electrical current, the time required for cooling per seizure, and the number of seizures per day, respectively. b was assumed to be 10 Ah. $\bar{I}$ is the sum of the electrical current supplied to the Peltier device, the pump, and the fan. The current for the Peltier device was 4.0 A based on section V-B, and those for the pump and fan were 0.2 and 0.3 A, respectively [26], [31]. Using these assumptions, the estimated average electrical current was approximately 5 A.

The cooling time for seizure suppression is estimated as the total time required for lowering and maintaining the brain surface temperature. In section V-B, the cooling period was determined to be ≤10 min, and maintenance of brain cooling should occur for about 10 min based on clinical experience at Yamaguchi University School of Medicine. That is, our system requires about 20 min of brain cooling for seizure suppression and consumes 1.5 Ah per cooling cycle. From this, we conclude that battery charging would be required every two or three seizures. The quartile frequencies of seizures with awareness impairment in temporal lobe epilepsy are 0.10, 0.16, and 0.47 /day [52], which indicates that charging or exchange of the battery would usually be required every few days. This frequency of required battery charging does not detract from the usability of the brain cooling system.

The intermittent brain cooling profile proposed by Yang *et al.* [42] has cooling for 30 s at intervals of 2 min.

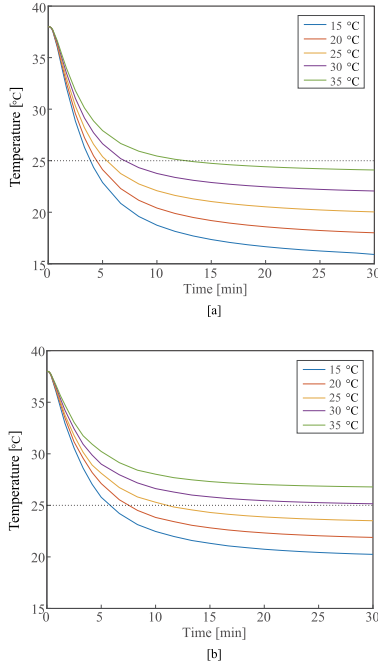


Fig. 12. Temporal change of average temperature at 2 mm below cooling device with different ambient temperatures. (a) W_{tis} is $5.2 \text{ kg}/(\text{m}^3 \cdot \text{s})$. (b) W_{tis} is $10.4 \text{ kg}/(\text{m}^3 \cdot \text{s})$.

TABLE VII
VALUES OF CONSTANTS

Constant	Value	Reason
$\Delta T_{\max}$	77 K	Specification sheet [29]
ν_s	Calculated by Eq. (5)	COMSOL Incorporated [43]
ρ_s	Calculated by Eq. (6)	COMSOL Incorporated [43]
ρ_{tis}	Shown in Table VI	ITIS Foundation [51]
c_{blood}	$4.2 \times 10^3 \text{ J}/(\text{kg} \cdot \text{K})$	Same as that of water
C_C	5.68 J/K	Lineykin and Yaakov [48]
C_H	5.68 J/K	Lineykin and Yaakov [48]
c_s	Calculated by Eq. (7)	COMSOL Incorporated [43]
c_{Ti}	$5.24 \times 10^2 \text{ J}/(\text{kg} \cdot \text{K})$	Haynes and Lide [54]
c_{tis}	Shown in Table VI	ITIS Foundation [51]
d_t	$4 \times 10^{-3} \text{ m}$	Easiness for handling
F	$6.7 \times 10^{-6} \text{ m}^3/\text{s}$	Experience
$I_{\max}$	8.5 A	Specification sheet [29]
k_s	Calculated by Eq. (8)	COMSOL Incorporated [43]
k_{Ti}	21.9 W/(m·K)	Haynes and Lide [54]
k_{tis}	Shown in Table VI	ITIS Foundation [51]
L_t	2 m	Double of distance from head to chest
R_C	0.46 K/W	The value on spec sheet [27] divided by 10, because heat sink is in contact with saline
R_H	0.5 K/W	Specification sheet [28]
T_a	298.15 K	Room temperature
T_{core}	311.15 K	Xu <i>et al.</i> [55]
$U_{\max}$	15.7 V	Specification sheet [29]
V_{tank}	$2.5 \times 10^{-5} \text{ m}^3$	Able to contain heat sink

This profile consumes 1.0 Ah per hour and the battery life is about 10 h. Thus, patients would need to charge or exchange the battery once or twice each day.

D. Effect of Ambient Temperature

The ambient temperature affects the cooling performance of the focal brain cooling system because the heat stored in the circulating saline is removed to the ambient air by the Peltier device. To evaluate the effect of the ambient temperature on the cooling performance, we simulated changes of the brain temperature with different ambient temperatures. Figure 12 shows

temporal changes of the average temperature at 2 mm below the cooling device at ambient temperatures of 15, 20, 25, 30, and 35 °C. The voltage for driving the Peltier device was fixed at 5 V and the initial saline temperature was 38 °C.

For $W_{tis} = 5.2 \text{ kg}/(\text{m}^3 \cdot \text{s})$, the average temperature at 2 mm below the cooling device reached the target of 25 °C within 15 min, even when the ambient temperature was 35 °C. This cooling is sufficient for seizure suppression. In contrast, with $W_{tis} = 10.4 \text{ kg}/(\text{m}^3 \cdot \text{s})$, the average temperature at 2 mm below the cooling device did not reach 25 °C after 30 min when the ambient temperature was 30 or 35 °C. This result indicates that it is difficult to suppress seizures when the ambient temperature is very high. Suppression of seizures at high ambient temperature requires an increased heat removal performance of the Peltier device, but this setting may require frequent battery exchanges.

VI. CONCLUSIONS

In this work, we developed mathematical models for a focal brain cooling system with recirculating coolant cooling (RCC) to achieve an optimal design. The model system consists of the head model, basic fluid dynamics equations, and a modified Pennes bioheat equation. To determine a parameter in the bioheat equation, an experiment was performed in a cat model. In addition, a pressure drop model of the relationship between the saline flow rate and pressure drop was constructed to reduce the computational burden of the CFD simulation.

The simulation results showed that a complex channel structure in the cooling device is desirable since it gives higher brain surface temperature uniformity and a lower pressure drop compared to a simple channel structure. A time <10 min was required for the average temperature at 2 mm below the cooling device to reach the target temperature of 25.0 °C, which is a short enough period for brain cooling after seizure prediction by an intracranial EEG-based algorithm. In addition, the frequency of battery charge in our system was calculated to be once every few days for most patients. Therefore, the focal brain cooling system under development is feasible for future treatment of epilepsy.

We plan to manufacture a prototype of the focal brain cooling system designed in this work and test it in animal experiments before clinical application. In addition, we will simulate various device sizes. A new seizure prediction algorithm will be constructed for determination of the timing for activation and deactivation of the cooling system, using EEG, brain temperature, intracranial pressure, and oxygen saturation and concentration of oxyhemoglobin in cerebral blood measured by skull-implanted multimodal sensors [53].

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